

Acme Support

AI customer support with human assistance

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Python / Streamlit / Azure AI Foundry / SQLite

The support problem

**Customers need answers.
Some requests need a person.**

Routine questions

Order status, delivery dates, and store policies.

Requests that need judgment

Refund exceptions, disputes, and explicit requests for human help.

The project at a glance

Customer chat

Customers enter order details, ask questions, and request a specialist.

Support console

Specialists view the transcript, send replies, and close tickets.

One saved conversation

Customer, AI, and human messages share a chronological history.

Application architecture

Interface	Streamlit customer chat and support console
Application	Python order lookup and conversation routing
AI service	Azure AI Foundry prompt agent
Persistence	SQLite orders, messages, tickets, and audit events

Customer experience

CUSTOMER

“Where is my order?”

ACME AI

“Your order has shipped.
Here is the estimated delivery date.”

Order context stays with the conversation after a refresh.

Human assistance

01 **Customer requests a human**

The app opens a ticket with the transcript and context.

02 **Specialist takes the conversation**

Human replies appear in the same customer chat.

03 **Customer chooses what happens next**

Continue with the specialist or switch back to AI.

04 **Specialist resolves the ticket**

The chat displays a closure notice and stops new messages.

The support workspace

Current customer first

The default view shows the latest active ticket.

Stable ticket selection

Sending a reply keeps the same case selected.

Context beside the reply

The specialist sees history, order details, and the handoff reason.

Visible resolution

Closing a ticket notifies the customer and closes the conversation.

Data and response boundaries

Stored order facts guide the answer

Lookup retrieves order status and delivery information from SQLite.
A supplied email must match the order's customer.

When the app cannot verify an answer

Missing orders or AI service failures create a human handoff.
Azure identity handles access to the AI service.

Live demonstration

Demo order

CS-1001

alex.morgan@example.com

- 01 Ask: “Where is my order?”
- 02 Request human assistance
- 03 Reply from the support console
- 04 Resolve and show the closure notice

Verification and improvements

5

offline quality checks passed

Workflow checks

Reply saves to the conversation
Selection survives status changes
Refresh restores the selected case
Latest active ticket opens by default

Validation covers demo behavior. Model accuracy and scale need separate evaluation.

Next development priorities

Identity and access

Customer sign-in and authenticated support-agent roles.

Team operations

Concurrent-agent assignment and a shared production database.

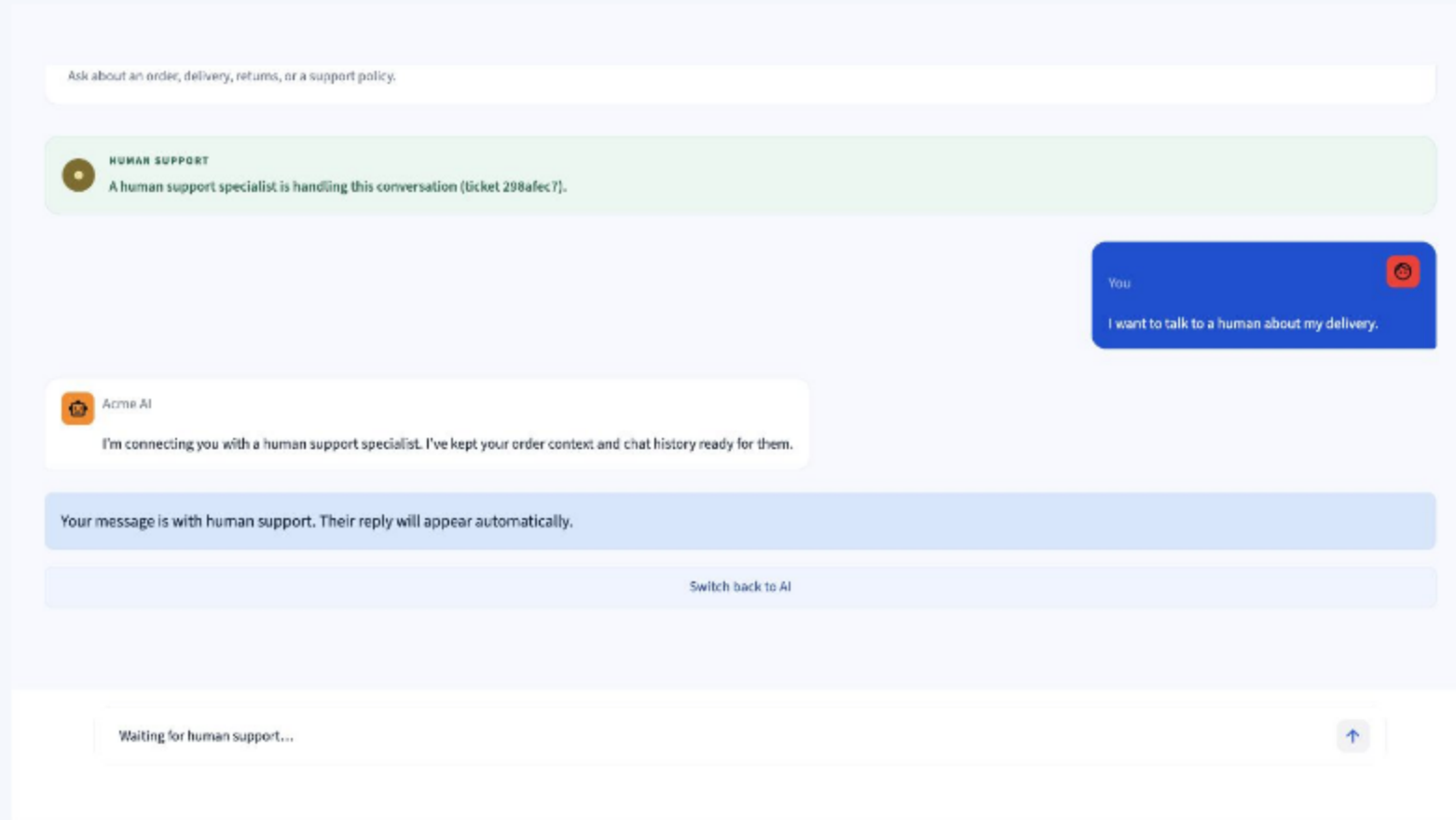
AI evaluation

Measure factual accuracy and strengthen multi-turn context.

Deployment

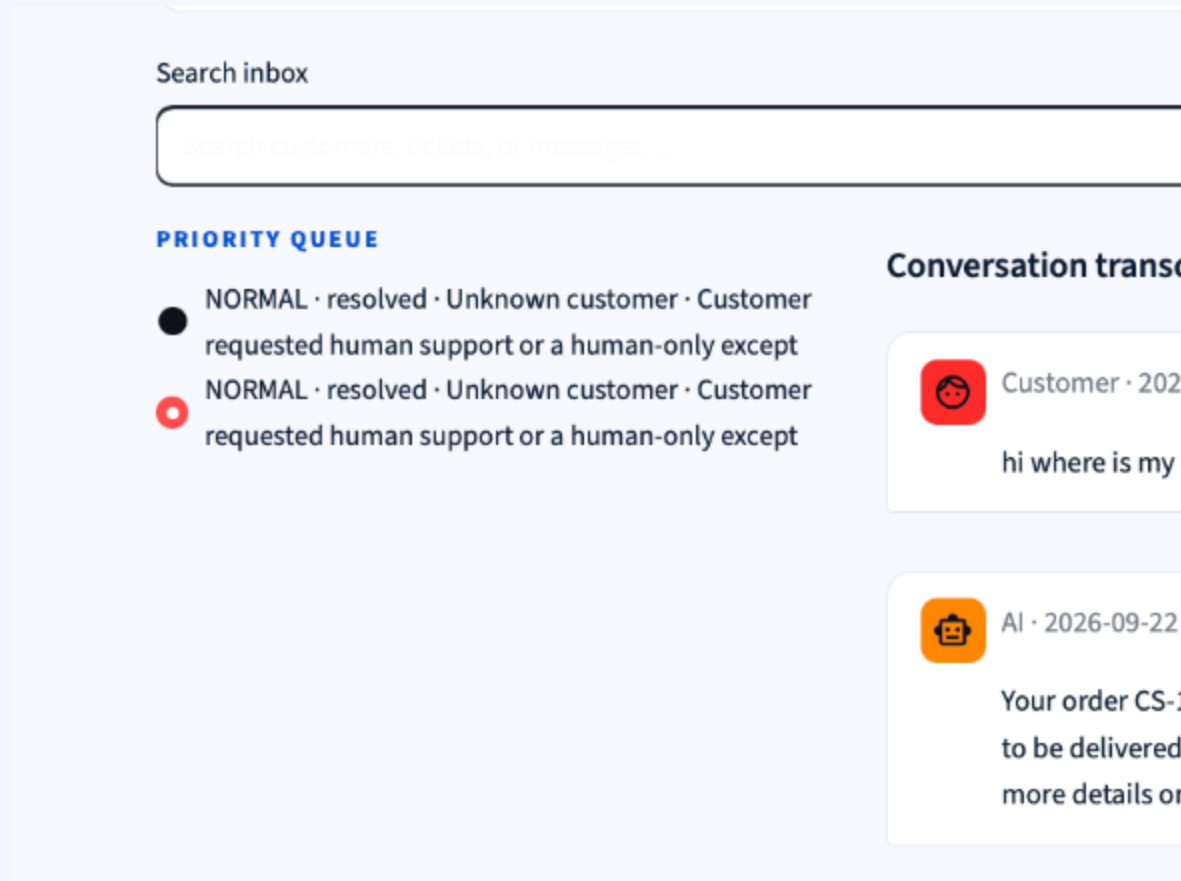
Hosted access, monitoring, backups, and usage controls.

Customer handoff in the running prototype



Customer request on the right. AI acknowledgement and handoff status on the left.

Support queue and conversation transcript



Earlier prototype screenshot: ticket selection beside the conversation transcript.

Project takeaway

A complete support conversation

AI handles routine questions.

A specialist takes over when the customer needs help.

Questions & live demo